

PID Simulation Construction Report

Version: 2.1

Date: 2026-08-04

Scope: Document the construction, assumptions, and validation of uavx-python/src/tests/test_pid_sim — the calibrated PID simulator used for the FW/MR tuning studies. **Supersedes v1 FW_SIMULATION_REPORT.md** (2026-07-26, “Aerodynamic Model v1”, qbar×S×C_ctrl physics without the 70%-deflection authority anchor). This is the authoritative construction reference.

v2.0 → v2.1 (2026-08-04): Added `is_roll_structurally_limited()` — aileron-less RudderElevatorAF frames have zero roll authority in the coupled model (roll routed to rudder, bank-and-yank cancelled by the yaw-angle hold). Roll step/gust criteria are non-scored for these frames, mirroring the existing yaw gate for rudder-less elevons. See §7.

1. Purpose

Simulate each airframe’s closed-loop step response (attitude, disturbance rejection, altitude hold, navigation) so PID gains can be tuned against physically meaningful authority before flight. The sim models the **full control path**: PID loops → FC mixer (DoServos/DoMotors) → aerodynamic/motor physics.

2. Architecture

```
run_tests_for_af(af_filename)
├─ get_params_for_af() → params, AirframeType, FW model idx, is_fw
├─ FW (descriptor present): simulate_axis_coupled() – 3-axis coupled aero
├─ FW/MR (legacy):         simulate_axis()         – single-axis
├─ Disturbance rejection:  simulate_rate_disturbance()
├─ Altitude hold:          simulate_alt_hold_fw() / simulate_alt_hold_mr()
├─ Navigation:             simulate_nav_fw() / simulate_nav_mr()
├─ Free-flight turbulence: simulate_freeflight() (FW only)
```

Active engine lives at `uavx-python/src/tests/test_pid_sim.py`. Copies under windows/, linux/, macos/ are stale snapshots — always run the uavx-python copy.

3. Parameter/Physics Sources

- **Params:** parsed from the `.af` file (`airframes.airframes.parse_af_file`) → {ParamIndex: raw_float} → `load_af_params()` names.
 - **FW physicals:** `.af` `PHYS_*` metadata merged over `FW_AIRFRAMES` defaults by `get_fw_descriptor()` (`PHYS_TO_DESCRIPTOR`). The `.af` is authoritative.
 - **MR physicals:** `PHYS_AUW_G`, `PHYS_ARM_MM`, `PHYS_MOTOR_COUNT`, `PHYS_MOTOR_THRUST_G` → hover/max thrust, arm length, inertia per frame (fallback `EM_*` emu.c defaults).
 - **Enums:** `protocol_enums.ParamIndex`, `AirframeType`, `AF_CATEGORY`, `FW_MODEL_IDX`, `AF_FILES`.
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4. FW Physics (simulate_axis_coupled, run_physics_fw)

Control torques

$L = C_{l_ail} \cdot \delta_a$ $C_{l_ail} = qbar \cdot S \cdot b \cdot CL_D_AIL$
 $M = C_{m_ele} \cdot \delta_e$ $C_{m_ele} = qbar \cdot S \cdot c \cdot CM_D_ELE$
 $N = C_{n_rud} \cdot \delta_r$ $C_{n_rud} = qbar \cdot S \cdot b \cdot CN_D_RUD$
 $N_{elevon} = adverse_yaw \cdot C_{l_ail} \cdot \delta_{elevon}$ (ElevonAF drag-differential)

Damping

quadratic: $C_{d_axis} \cdot r \cdot |r|$ (PHYS_*_DAMP)
linear: $C_{lin_axis} \cdot |r|$ (*_damp_lin)

Stability & coupling

- Pitch/yaw static stability: $\pm stab_coeff \cdot qbar \cdot S \cdot ref \cdot angle$
- Adverse yaw: $N_{adverse} = adverse_yaw \cdot L_{ctrl}$
- Dihedral: $L_{dihedral} = dihedral_coeff \cdot r_{yaw}$
- Servo lag: first-order servo_tau on all surface channels

Rate ceiling

_compute_fw_max_rates() solves the quadratic damping balance at full deflection ($\approx 1.195 \times MAX_*_RATE$). The sim clamps rates to this ceiling; **MAX_*_RATE is the 70%-deflection steady-state rate, not the clamp.**

Mixing (FC DoServos/DoMotors)

Efforts R/I/P/YI route per AF type — see §6 of FW_Control_Authority_Calibration.md. Elevon, Rudder/Elevator, Delta/Aileron/Spoileron paths all modeled; feedforwards FW_ROLL_PITCH_FF, FW_AILERON_RUDDER_MIX applied.

5. MR Physics (run_physics_mr, simulate_axis)

- Torque from motor differential thrust: $\tau = kT \cdot effort$, $kT = max_thrust \cdot 0.25 \cdot arm$.
 - Axis damping: Roll 0.015, Pitch 0.03, Yaw 0.05 (quadratic, sign-opposed).
 - Motor lag: first-order EM_MOTOR_TAU (0.10 s).
 - Inertia from arm/mass per frame (shape factor by motor count: 4→0.55, 6→0.40, 8→0.30).
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6. Outer Loops

Altitude Hold

- **MR** (simulate_alt_hold_mr): position PI → desired ROC → velocity PI → throttle comp. Fast tilt response.
- **FW** (simulate_alt_hold_fw): position PI (ALT_POS_KP/KI) → ROC setpoint → velocity PI (ALT_ROC_KP, UNUSED_ALT_VEL_KI) → climb with climb_tau = 2.0 s aero lag. Airspeed-bleed coupling.

Navigation

- **MR** (simulate_nav_mr): cross-track error → NAV_POS_KP/KI → velocity → bank (NAV_VEL_KP → $g \cdot \tan(\text{bank})$) — fast (~1 s).
 - **FW** (simulate_nav_fw): cross-track → NAV_POS_KP/KI → bank-angle command → coordinated turn heading_rate = $g \cdot \tan(\text{bank})/V$ — slower (bank_tau = 1.5 s).
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7. Criteria

Test	Roll	Pitch	Yaw
MR attitude (CRITERIA)	rise≤1.0s, OS≤10%, settle≤2.5s, err≤1°	same	rise≤3s, OS≤15%, settle≤5s, err≤2°
FW attitude (FW_CRITERIA)	rise≤4s, OS≤25%, settle≤8s, err≤5°	rise≤5s, OS≤30%, settle≤12s, err≤15°	rise≤5s, OS≤25%, settle≤10s, err≤5°
Disturbance (DIST_CRITERIA)	peak≤15°/s, settle≤1.5s, IE≤10°	peak≤12°/s, IE≤8°	peak≤12°/s, IE≤10°
AH (MR/FW)	AH_CRITERIA_MR rise≤2s OS≤20% settle≤5s err≤1°	AH_CRITERIA_FW rise≤6s OS≤15% settle≤8s err≤2° (climb-tau floor ~4.4s)	—
Nav (MR/FW)	rise≤2s/3s, OS≤20%/15%, settle≤5s, err≤1°/2°	—	—

Structural exclusions: rudderless elevons (is_yaw_structurally_limited) — yaw step/gust non-scored; aileron-less RudderElevatorAF frames (is_roll_structurally_limited) — roll step/gust non-scored.

8. Free-Flight (FW)

Open-loop 8s turbulence response with zero surface deflection — measures **intrinsic** stability (dihedral, static margin, damping) without PID. Reports roll/pitch/yaw peak excursions. A large roll divergence (e.g. Shadow 1286°) is a **structural** characteristic of low-dihedral flying wings, not a tuning failure.

9. Known Gaps / Caveats

1. **Rudder-only roll** (is_roll_structurally_limited, v2.1): CL_D_AIL = 0 → zero aileron authority → roll ceiling 0. Bank-and-yank roll authority under-modeled (rudder produces yaw → dihedral couples to roll), so roll is **non-scored** for these frames rather than reported as an unreachable FAIL.
2. **Elevon yaw:** drag-differential authority is approximate (~1-3° hold). No differential-motor yaw (all FW frames single-motor; RudderMotorFF path exists in FC but no multi-motor FW frame in the set).

3. **Airspeed coupling:** AH/Nav FW use fixed cruise speed; no throttle-induced pitch coupling beyond `climb_tau`.
 4. **DT/EM mismatch:** windows/, linux/, macos/ sim copies are stale — the authoritative file is `uavx-python/src/tests/test_pid_sim.py`.
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10. Files Modified (this calibration)

- `uavx-python/src/tests/test_pid_sim.py` — `get_fw_descriptor`, `load_af_physicals`, `quadratic_compute_fw_max_rates`, `PHYS_T0_DESCRIPTOR`, `C_d_roll` from `PHYS_ROLL_DAMP` (3 call sites), elevon yaw mixing, criteria relax, **v2.1** `is_roll_structurally_limited` + Roll gating in `run_tests_for_af`.
- 12 FW .af files — `PHYS_*` aero block + `RATE_KP = 0.7/MAX_*_RATE`.
- Reports: `wiki/FW_Control_Authority_Calibration.md` (this series), `wiki/tuning_study.md`.